LEVOFLOXACIN

Tendon rupture risk

Once daily

CHEMICAL NAME (S)-9-fluoro-2,3-dihydro-3-methyl-10-(4-methylpiperazin-1-yl)-7-oxo-7H-pyrido[1,2,3-de]-1,4-benzoxazine-6-carboxylic acid

CLASS Respiratory fluoroquinolone (third generation)

TRADE NAMES Tavanic · Levaquin · Cravit · Levoxin

THERAPEUTIC USES / INDICATIONS

- Community-acquired pneumonia; acute exacerbation of COPD; sinusitis (when alternatives unsuitable).
- Complicated UTI, pyelonephritis, chronic prostatitis.
- Skin infections; inhalational anthrax; plague.
- Multidrug-resistant TB (second-line); *H. pylori* salvage therapy.

ADMINISTRATION

- **PO** once daily; separate from antacids, iron, zinc, calcium by 2 h.
- **IV** infusion: 500 mg over at least 60 min (250 mg over 30 min).
- Oral and IV doses are equivalent.

DOSAGES

- **Adult:** 500–750 mg once daily.
- **CAP:** 500 mg once or twice daily, or 750 mg once daily for 5 days.
- **UTI:** 250–750 mg once daily.
- **CrCl 20–49:** 250–500 mg every 24–48 h per indication; **<20:** further reduction.

PHARMACOKINETICS

- **Peak:** 1–2 h. Bioavailability ~99%.
- Minimal metabolism.
- Excreted unchanged in urine. **t½:** 6–8 h.

PHARMACODYNAMICS

- Inhibits **DNA gyrase and topoisomerase IV** → bactericidal.
- Enhanced Gram-positive (pneumococcal) and atypical cover compared with ciprofloxacin; good Gram-negative activity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea.
- Headache, dizziness, insomnia.
- Photosensitivity.

ADVERSE EFFECTS (SERIOUS)

- Tendon rupture; peripheral neuropathy.
- QT prolongation; aortic aneurysm.
- Hypo- or hyperglycaemia (diabetics).
- Seizures, psychiatric reactions; *C. difficile* colitis.

CONTRAINDICATIONS & CAUTIONS

- Fluoroquinolone hypersensitivity; previous FQ tendon injury.
- Epilepsy; children and adolescents; pregnancy and breastfeeding.
- **Caution:** long QT, elderly on steroids, myasthenia gravis, diabetes.

KEY INTERACTIONS

- **Antacids, Fe, Zn, Ca:** ↓ absorption.
- **QT-prolonging drugs:** additive.
- **Warfarin:** ↑ INR.
- **Steroids:** ↑ tendon risk; **antidiabetics:** dysglycaemia.

NURSING MANAGEMENT

- Check renal function for dosing.
- Monitor glucose in diabetics; ECG in at-risk patients.
- Assess for tendon pain, neuropathy, confusion.
- IV: infuse over 60 min.
- Monitor respiratory status in pneumonia.

PATIENT EDUCATION

- Report tendon pain, tingling, palpitations, mood changes.
- Avoid sun exposure.
- Separate from minerals and antacids.

KEY POINTS

- Respiratory FQ — pneumococcal + atypical cover.
- Tendon, QT, glucose problems.
- Once daily; oral = IV.
- Renal dose adjustment.

SOURCES CHECKED **[170]** Levofloxacin 250mg tablets (UK SmPC) **[171]** Levofloxacin 5 mg/ml solution for infusion (UK SmPC) **[172]** Levofloxacin, oral (US label) **[173]** Levofloxacin, IV (US label)

Full references p. 432–457